PORTI LAB 1

🔐 Azure RBAC Walkthrough: Users + Groups + Group-Based RBAC (Portal, PowerShell, Bash)

What I built

A proof-of-concept showing how to:

1. Create users and groups in **Microsoft Entra ID**
2. Assign users to groups
3. Assign an Azure RBAC role to a **group** (not a user)
4. Verify effective access at a **resource group scope** (`FortisAid-Lab`, East US)

Built to demonstrates **least privilege** and reduces permission sprawl by using group-based access control.

Step 0: Target Architecture (What I intended)

Users

- Emmanuel Tobi (Senior Admin)
- Tobi Babalola (Junior Admin)
- Eneattah Grace (Service Desk)

Groups

- Senior Admins
- Junior Admins
- Service Desk

RBAC

- Assign **Virtual Machine Contributor** → to **Service Desk group**
- Scope: **Resource Group** `FortisAid-Lab`

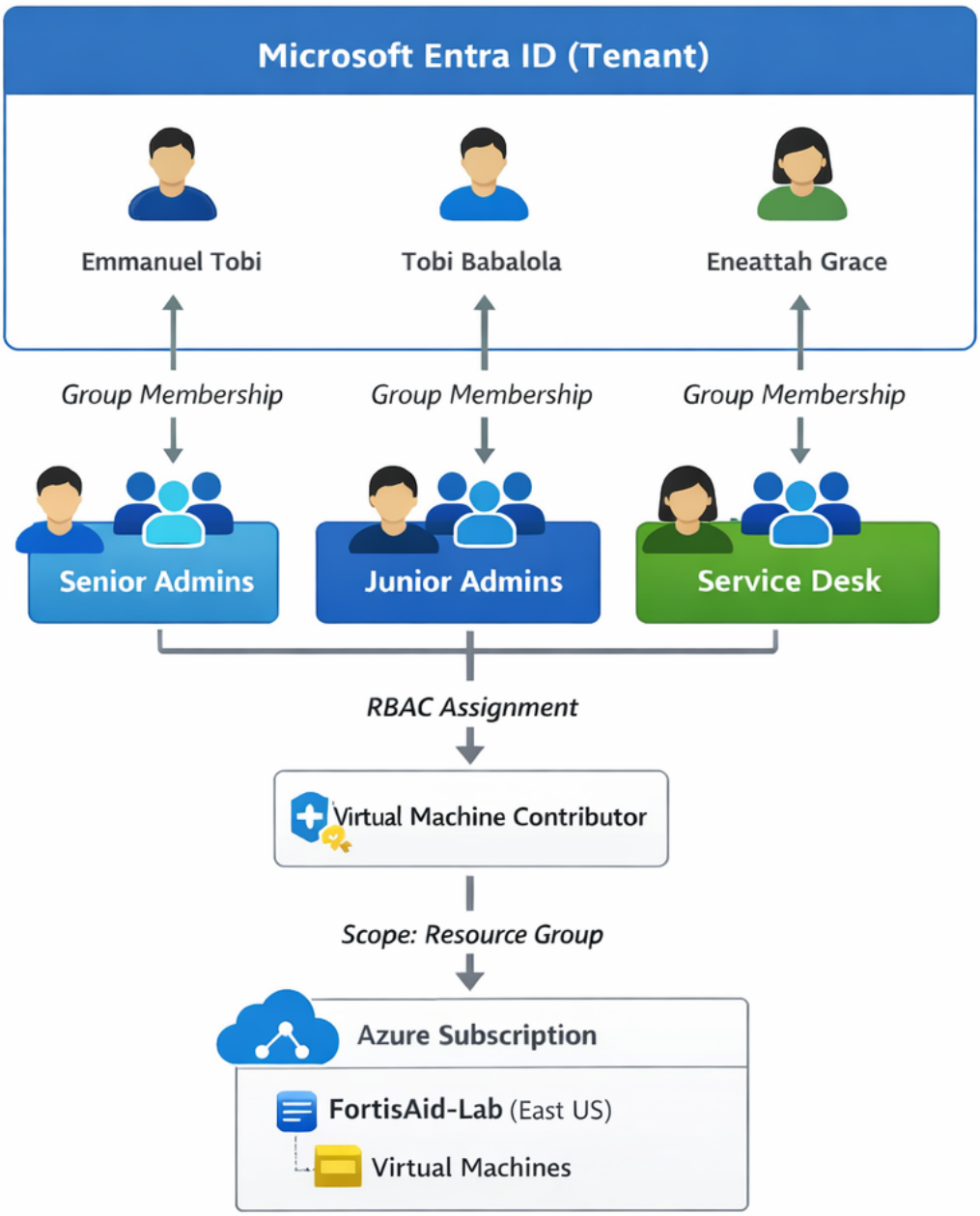


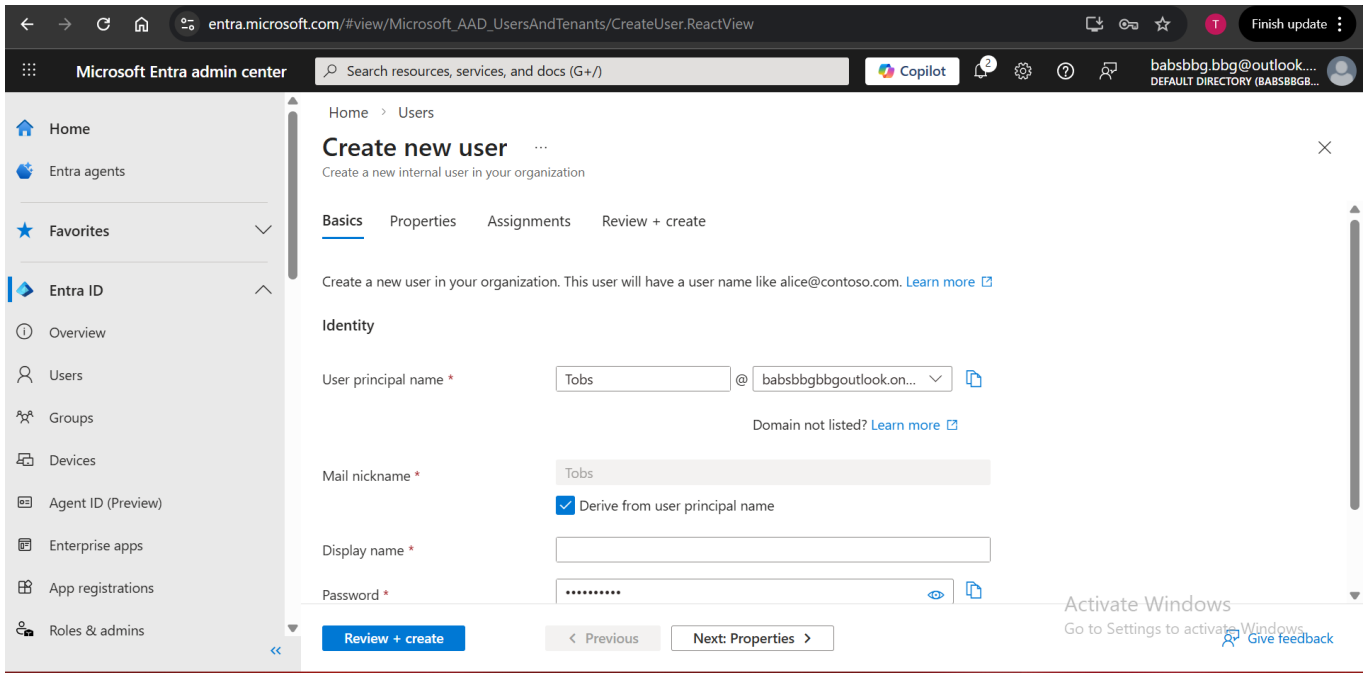
Figure: Azure RBAC architecture showing group-based access control using Microsoft Entra ID, where the Service Desk group inherits the Virtual Machine Contributor *role scoped to the FortisAid-Lab resource group.*

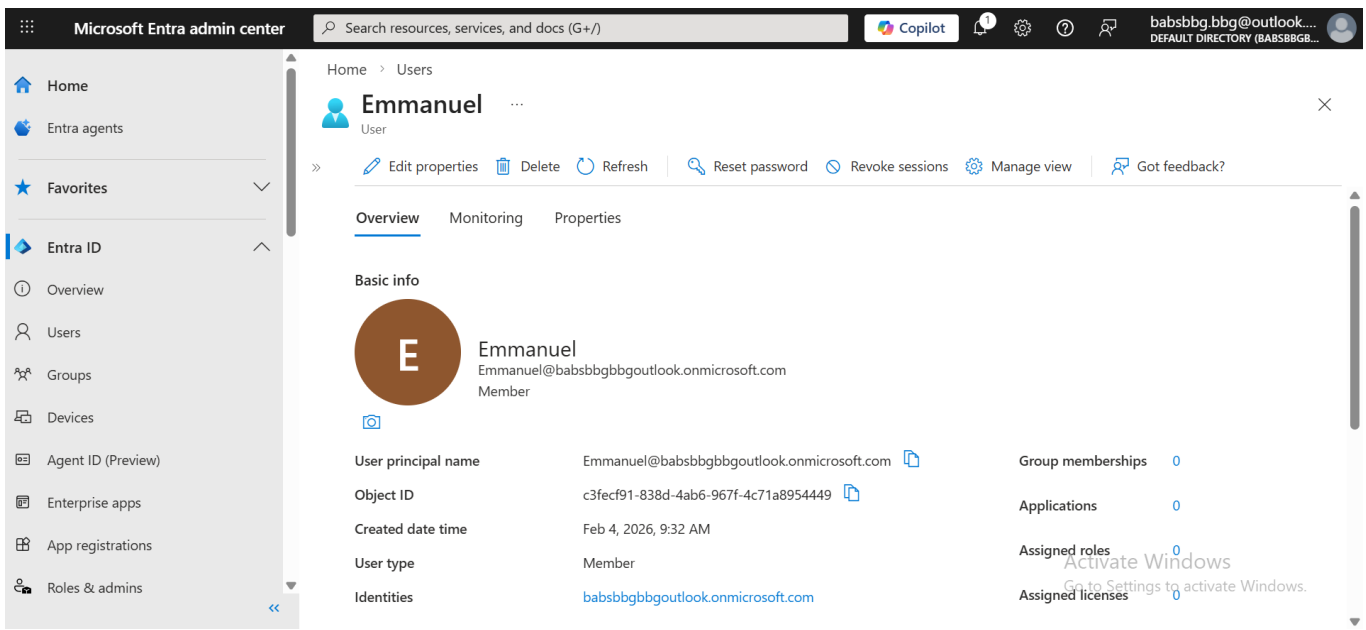
Step 1: Create Senior Admin user (Portal) — Emmanuel Tobi

Steps (Azure Portal → Entra ID)

1. Sign in to `https://portal.azure.com`
2. Search **Microsoft Entra ID**
3. Go to **Manage** → **Users**
4. Click **+ New user** → **Create new user**
5. Fill in:
   - **User name:** `emmanuel.tobi@<tenant>.onmicrosoft.com`
   - **Name:** `Emmanuel Tobi`
   - **Password:** Auto-generate (save it)
   - **Account enabled:** Yes
6. Click **Create**

✅ **Verification:**
Microsoft Entra admin center → Users → **Emmanuel Tobi** appears.





Step 2: Create Junior Admin user (PowerShell) — Tobi Babalola

Why PowerShell here?

Shows **repeatable / auditable identity automation**, suitable for enterprise environments.

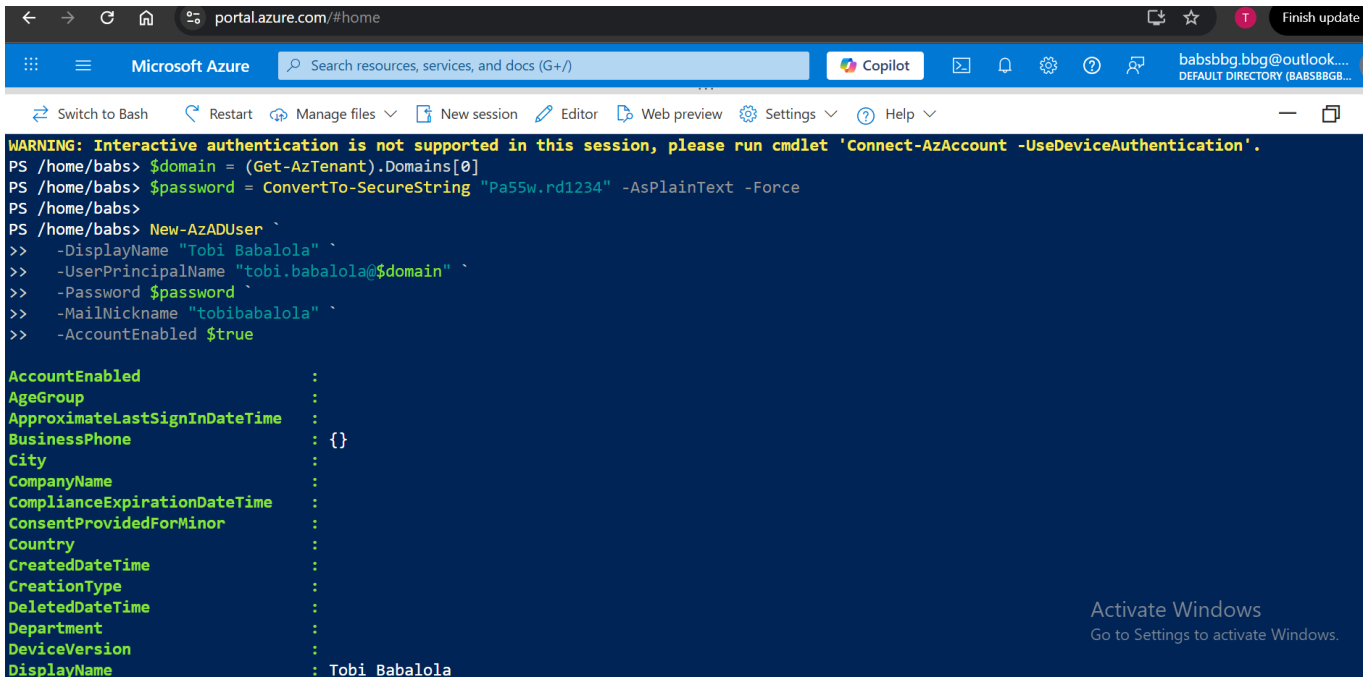
Actions (Cloud Shell → PowerShell)

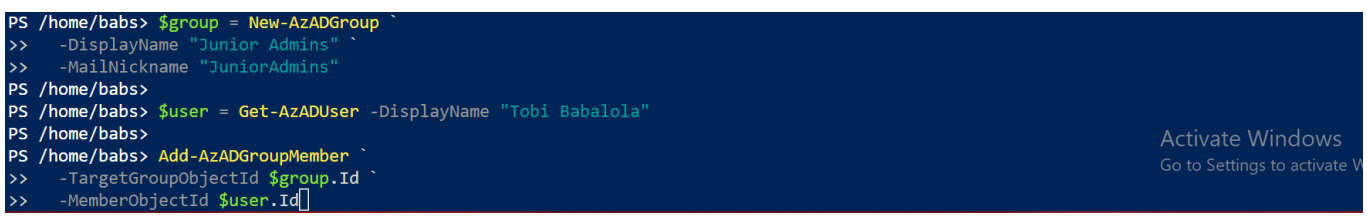
```
Connect-AzAccount
$domain = (Get-AzTenant).Domains[0]
```

```
$password = ConvertTo-SecureString "Pa55w.rd1234" -AsPlainText -Force

New-AzADUser `
  -DisplayName "Tobi Babalola" `
  -UserPrincipalName "tobi.babalola@$domain" `
  -Password $password `
  -MailNickname "tobibabalola" `
  -AccountEnabled $true
```

✅ **Verification:**

```
Get-AzADUser -DisplayName "Tobi Babalola"
```





Step 3: Create Service Desk user (Bash / Azure CLI) — Eneattah Grace

Why Bash / CLI here?

Shows cross-platform automation and the same approach used inside CI/CD pipelines.

Important note (real-world issue I handled)

I hit a UPN domain error until I used a **verified tenant domain**. The verified domain I used was:

`babsbbgbbgoutlook.onmicrosoft.com`

Actions (Cloud Shell → Bash)

```
DOMAIN="babsbbgbbgoutlook.onmicrosoft.com"

az ad user create \
  --display-name "Eneattah Grace" \
  --user-principal-name eneattah.grace@$DOMAIN \
  --password "Pa55w.rd1234"
```

✅ **Verification:**

```
az ad user show --id eneattah.grace@$DOMAIN --query displayName -o tsv
```

Expected output:

`Eneattah Grace`

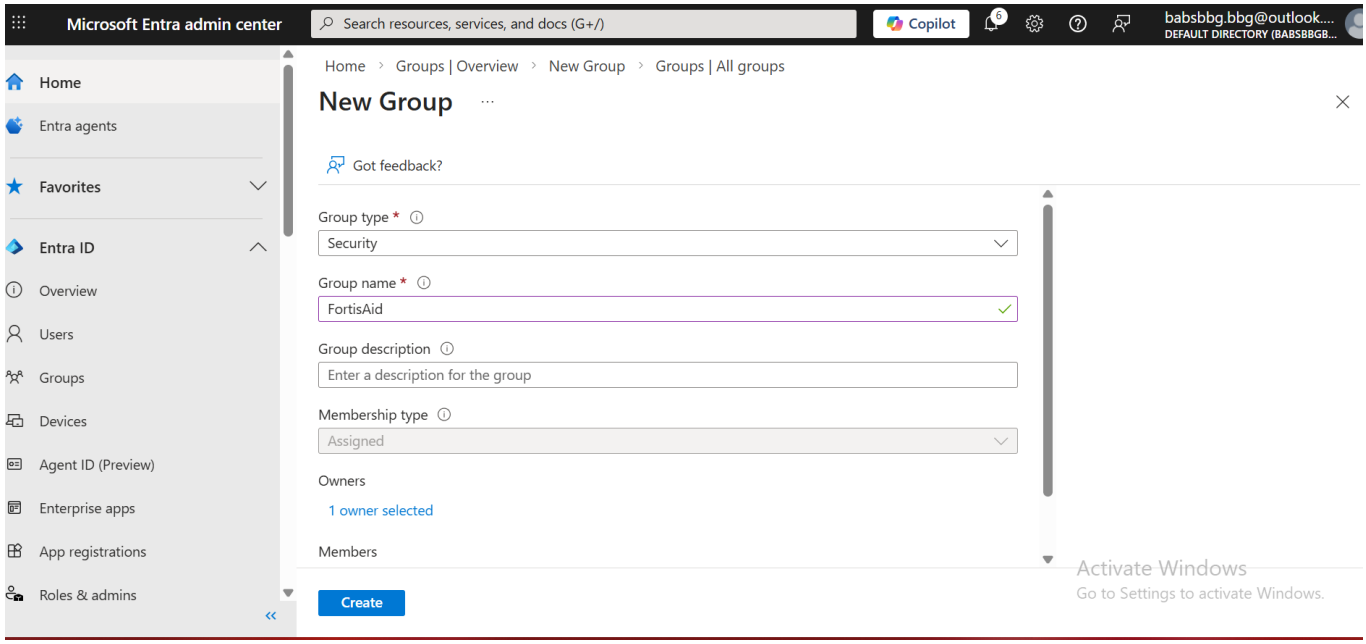
```
babs [ ~ ]$ DOMAIN="babsbbgbbgoutlook.onmicrosoft.com"

az ad user create \
  --display-name "Eneattah Grace" \
  --user-principal-name eneattah.grace@$DOMAIN \
  --password "Pa55w.rd1234"
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#users/$entity",
  "businessPhones": [],
  "displayName": "Eneattah Grace",
  "givenName": null,
  "id": "e81194e0-20e8-4b87-bd52-6731a2f2b143",
  "jobTitle": null,
  "mail": null,
  "mobilePhone": null,
  "officeLocation": null,
  "preferredLanguage": null,
  "surname": null,
  "userPrincipalName": "eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com"
}
babs [ ~ ]$ az ad user show --id eneattah.grace@$DOMAIN --query displayName -o tsv
Eneattah Grace
babs [ ~ ]$
```

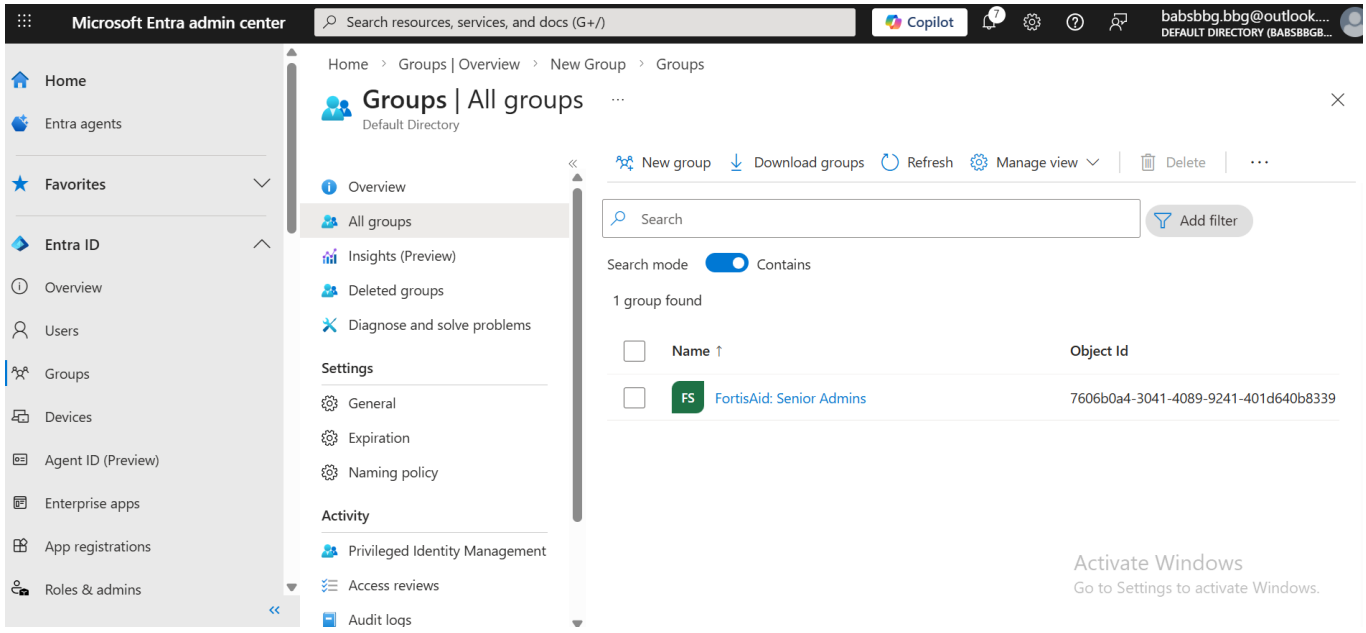
Step 4: Create groups and add members (Entra ID)

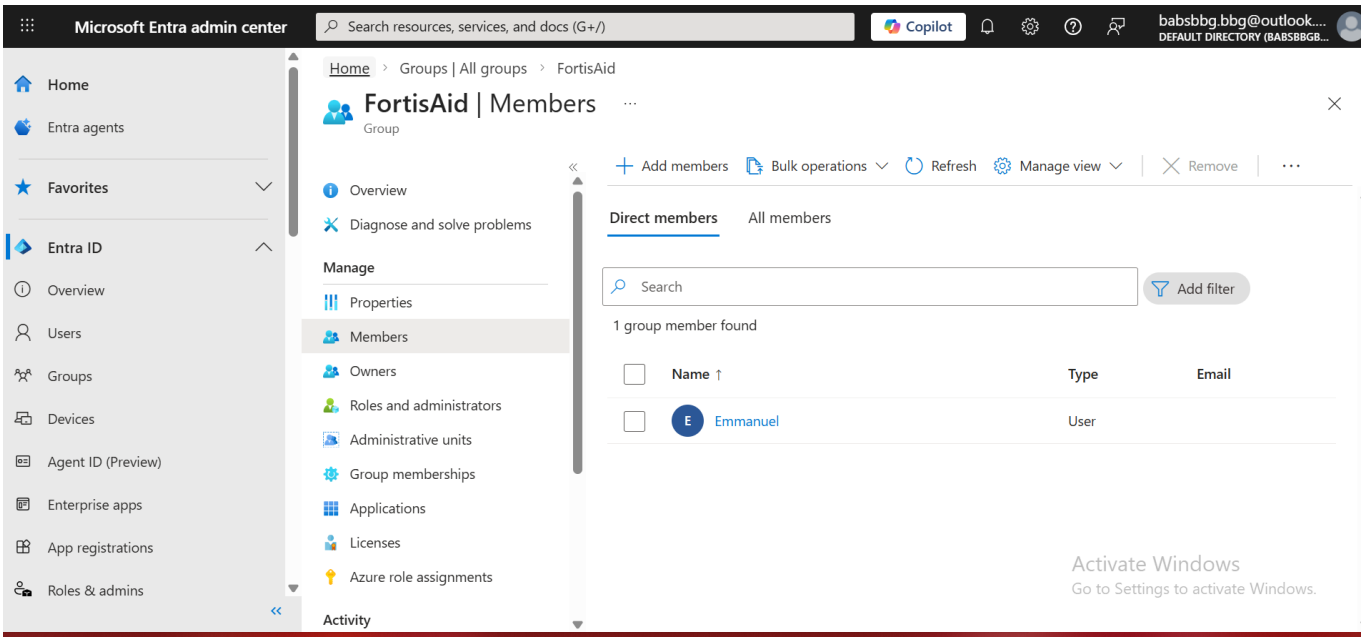
4A) Create Senior Admins group + add Emmanuel Tobi (Portal)

1. Entra ID → **Groups** → **+ New group**
2. Type: **Security**
3. Name: **Senior Admins**
4. Membership type: **Assigned**
5. Add member: **Emmanuel Tobi**

6. Create

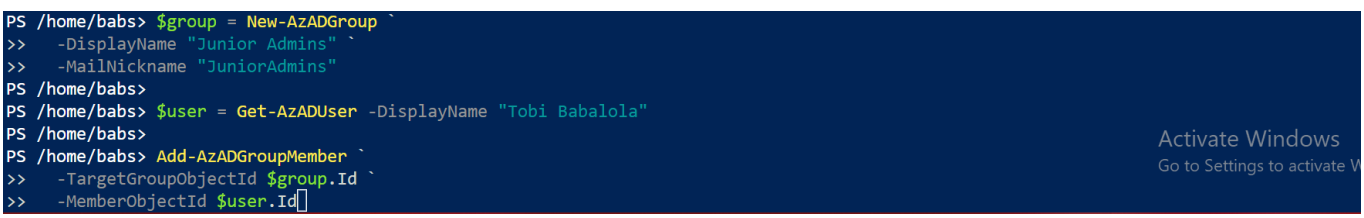

Microsoft Entra admin center
Search resources, services, and docs (G+/)
Copilot
babsbbg.bbg@outlook....
DEFAULT DIRECTORY (BABSBBGB...
Home
Entra agents
Favorites
Entra ID
Overview
Users
Groups
Devices
Agent ID (Preview)
Enterprise apps
App registrations
Roles & admins
Home > Groups | Overview > New Group > Groups | All groups
New Group
Got feedback?
Group type *
Security
Group name *
FortisAid
Group description
Enter a description for the group
Membership type
Assigned
Owners
1 owner selected
Members
Create
Activate Windows
Go to Settings to activate Windows.


Microsoft Entra admin center
Search resources, services, and docs (G+/)
Copilot
babsbbg.bbg@outlook....
DEFAULT DIRECTORY (BABSBBGB...
Home
Entra agents
Favorites
Entra ID
Overview
Users
Groups
Devices
Agent ID (Preview)
Enterprise apps
App registrations
Roles & admins
Home > Groups | Overview > New Group > Groups
Groups | All groups
Default Directory
Overview
All groups
Insights (Preview)
Deleted groups
Diagnose and solve problems
Settings
General
Expiration
Naming policy
Activity
Privileged Identity Management
Access reviews
Audit logs
New group
Download groups
Refresh
Manage view
Delete
Search
Add filter
Search mode
Contains
1 group found
Name
Object Id
FS
FortisAid: Senior Admins
7606b0a4-3041-4089-9241-401d640b8339
Activate Windows
Go to Settings to activate Windows.



4B) Create Junior Admins group + add Tobi Babalola (PowerShell)

```
$group = New-AzADGroup `
  -DisplayName "Junior Admins" `
  -MailNickname "JuniorAdmins"

$user = Get-AzADUser -DisplayName "Tobi Babalola"

Add-AzADGroupMember `
  -TargetGroupObjectId $group.Id `
  -MemberObjectId $user.Id
```



4C) Create Service Desk group + add Eneattah Grace (Bash)

```
az ad group create \
  --display-name "Service Desk" \
  --mail-nickname servicedesk

USER_ID=$(az ad user show \
  --id eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com \
  --query id -o tsv)
```

```
GROUP_ID=$(az ad group show \
  --group "Service Desk" \
  --query id -o tsv)

az ad group member add \
  --group $GROUP_ID \
  --member-id $USER_ID
```

✅ Verification:

```
az ad group member list \
  --group "Service Desk" \
  --query "[].displayName" -o table
```

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help

```
babs [ ~ ]$ az ad group create \
  --display-name "Service Desk" \
  --mail-nickname servicedesk
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#groups/$entity",
  "classification": null,
  "createdDateTime": "2026-02-04T09:00:40Z",
  "creationOptions": [],
  "deletedDateTime": null,
  "description": null,
  "displayName": "Service Desk",
  "expirationDateTime": null,
  "groupTypes": [],
  "id": "e46addca-dbf0-4508-b81a-4ab5fe7770a7",
  "isAssignableToRole": null,
  "mail": null,
  "mailEnabled": false,
  "mailNickname": "servicedesk",
  "membershipRule": null,
  "membershipRuleProcessingState": null,
  "onPremisesDomainName": null,
  "onPremisesLastSyncDateTime": null,
  "onPremisesNetBiosName": null,
  "onPremisesProvisioningErrors": [],
  "onPremisesSamAccountName": null,
  "onPremisesSecurityIdentifier": null,
  "onPremisesSyncEnabled": null,
  "preferredDataLocation": null,
  "preferredLanguage": null,
```

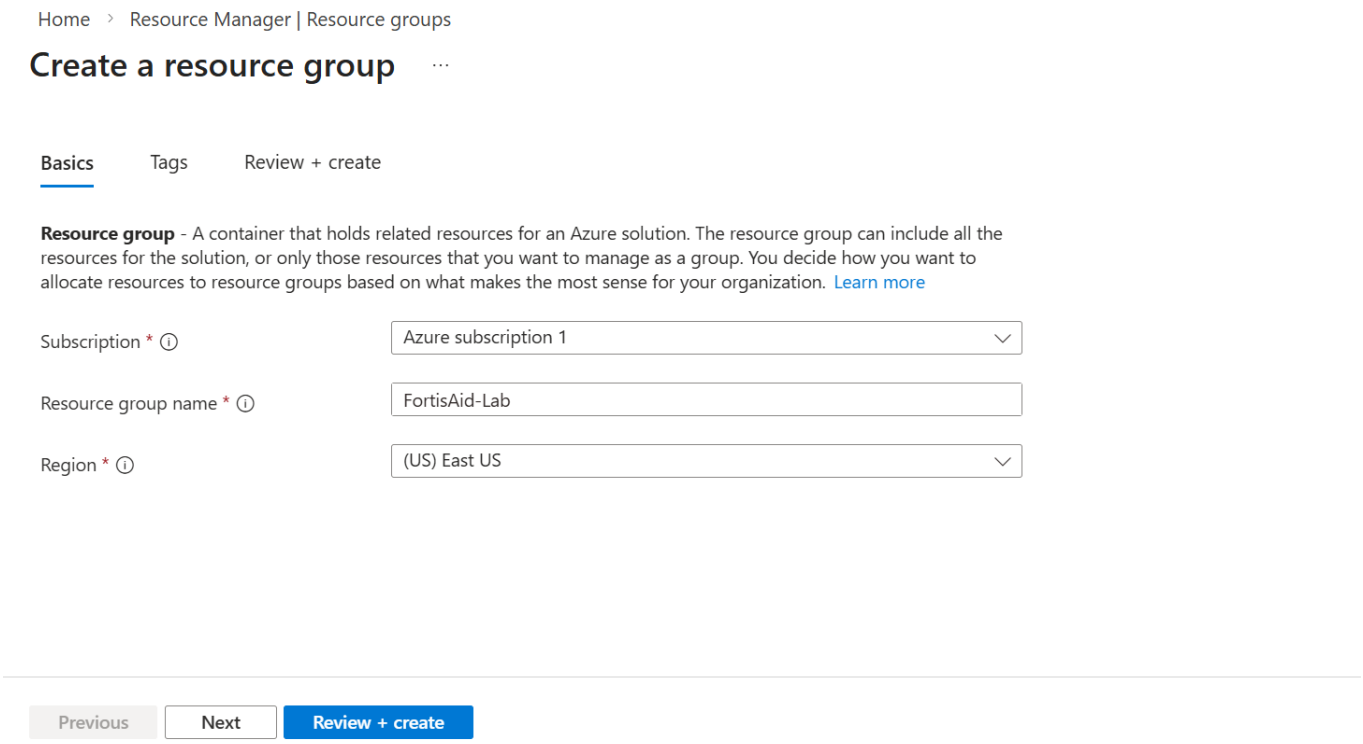
```
babs [ ~ ]$ USER_ID=$(az ad user show \
  --id eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com \
  --query id -o tsv)

GROUP_ID=$(az ad group show \
  --group "Service Desk" \
  --query id -o tsv)
babs [ ~ ]$ az ad group member add \
  --group $GROUP_ID \
  --member-id $USER_ID
babs [ ~ ]$ az ad group member list \
  --group "Service Desk" \
  --query "[].displayName" -o table
Result
--------------
Eneattah Grace
babs [ ~ ]$
```

Step 5: Assign RBAC Role to the Service Desk group (Scope: FortisAid-Lab)

Goal

Service Desk can manage VMs but can't do subscription-wide admin tasks.

1. Azure Portal → Resource groups → **FortisAid-Lab**
2. **Access control (IAM) → + Add → Add role assignment**
3. Role: **Virtual Machine Contributor**
4. Assign access to: **User, group, or service principal**
5. Select: **Service Desk**
6. **Review + assign**



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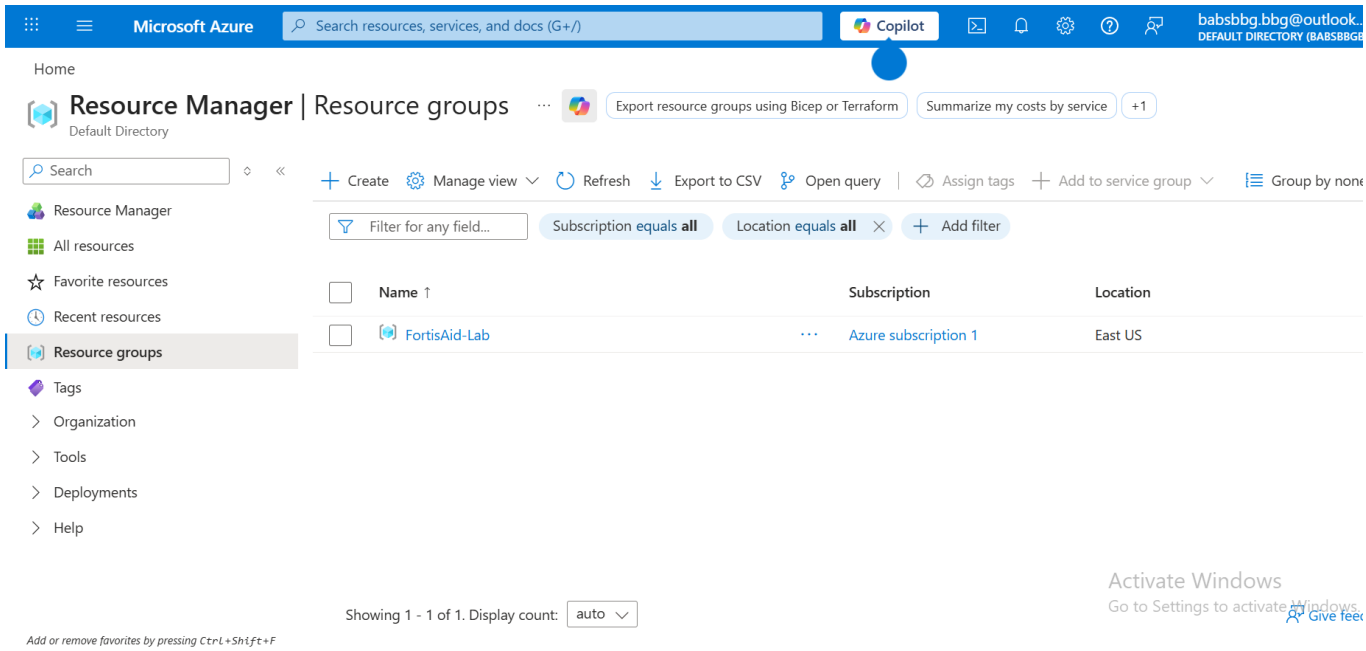
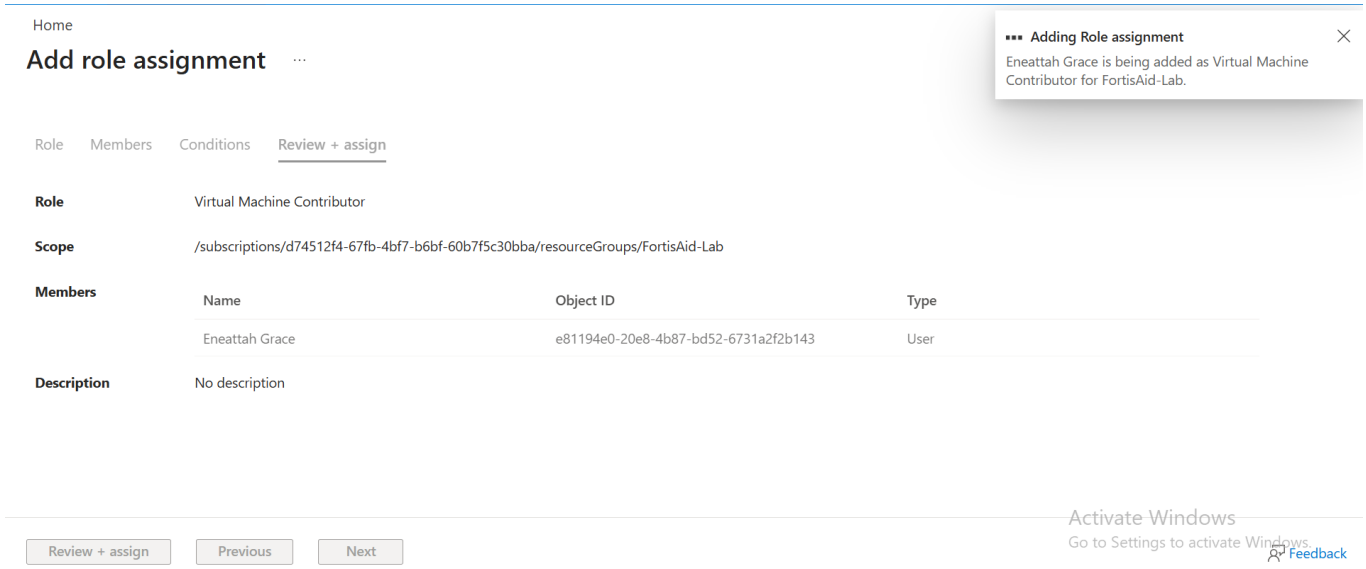
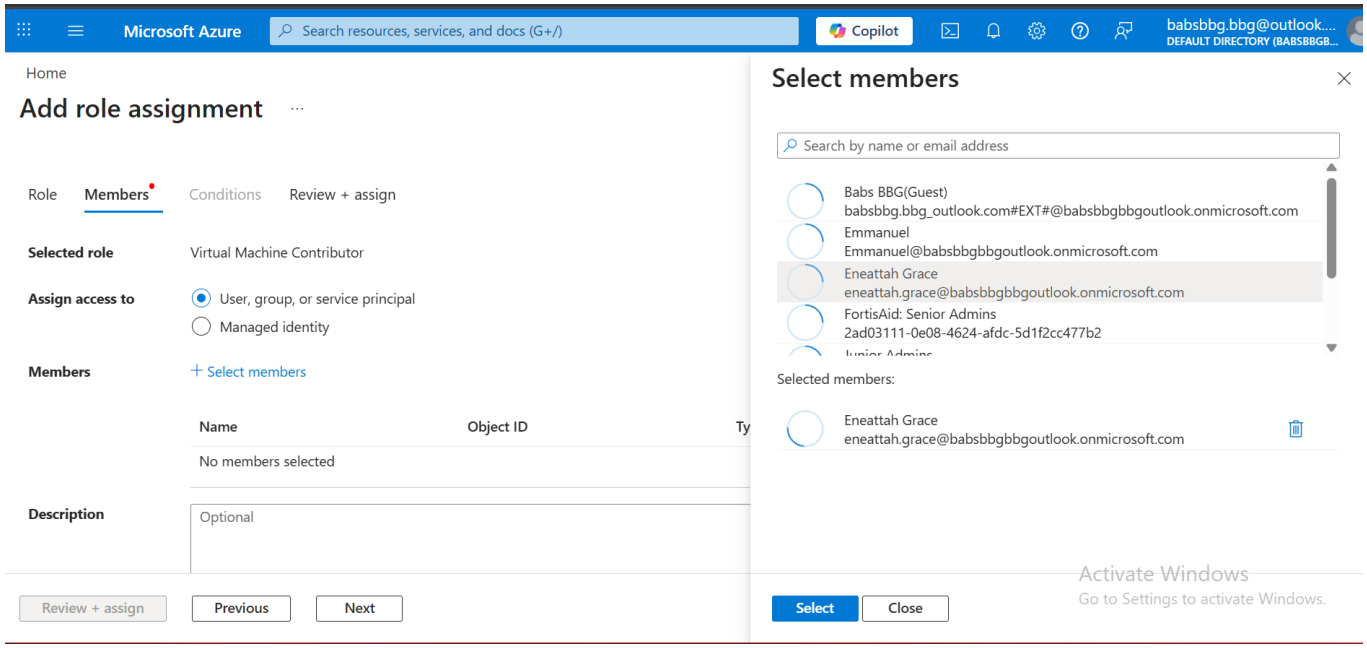




Step 6: Verify effective permissions (Critical)

1. FortisAid-Lab → **Access control (IAM)**
2. **Check access**
3. Search: **Eneattah Grace**
4. Confirm:
   - **Role:** Virtual Machine Contributor
   - **Assignment source:** Service Desk group
   - **Scope:** FortisAid-Lab

This proves the user inherits privileges through group membership.



Why this is security-relevant

- **Least privilege:** Role scoped to a resource group, not the whole subscription
- **Scalable access:** Add/remove members from groups without changing RBAC
- **Audit-friendly:** Group assignments simplify access reviews and compliance checks
- **Zero trust aligned:** No direct user privilege grants, controlled inheritance only

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⚠️ Common Pitfalls Encountered & Resolve

Pitfall 1: Invalid userPrincipalName domain (CLI)

Issue:
User creation failed with:

> "The domain portion of the userPrincipalName property is invalid."

Root cause:
Azure Entra ID only allows user creation using **verified tenant domains**.

Resolution:
Used the tenant's verified domain:

```
babsbbgbbgoutlook.onmicrosoft.com
```

Security takeaway:
Enforcing verified domains prevents identity spoofing and untrusted namespaces.

Pitfall 2: `New-AzADGroup` parameter mismatch (PowerShell)

Issue:
PowerShell returned:

> "A positional parameter cannot be found that accepts argument 'True'."

Root cause:
`New-AzADGroup` does **not** accept `-SecurityEnabled $true`.
Security groups are created by default.

Resolution:
Removed the unsupported parameter.

```
New-AzADGroup -DisplayName "Junior Admins" -MailNickname "JuniorAdmins"
```

Security takeaway:
Understanding cmdlet behavior prevents automation errors and misconfiguration.

Pitfall 3: `az ad domain` command unavailable

Issue:
`az ad domain list` not recognized in Cloud Shell.

Root cause:
Azure CLI has moved some Entra ID functionality behind Microsoft Graph.

Resolution:
Verified domains via:

- Microsoft Entra admin center
- Microsoft Graph (via `az rest`)

Security takeaway:
Identity operations increasingly rely on Graph APIs rather than legacy CLI surfaces.

Pitfall 4: Confusing “az login” behavior in Cloud Shell

Issue:
`az login` prompted device authentication unexpectedly.

Root cause:
Cloud Shell is **already authenticated** using the portal session.

Resolution:
Skipped `az login` and used `az account show` to confirm context.

Security takeaway:
Implicit authentication reduces credential exposure in managed shells.

📄 Final Portfolio Summary

> This walkthrough demonstrates a secure, enterprise-aligned approach to Azure RBAC using Microsoft Entra ID. By combining portal-based setup with PowerShell and Azure CLI automation, and validating effective access through group inheritance, the project reflects real-world cloud security practices rather than tutorial-driven configuration.